

Task on RDS

1. Create MariaDB DB on EC2.

- Created new instance in Ec2

☐	Name ↗	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm stat
☐	Maria-DB	i-0f39838ae4d0a7c94	Running 🔍 🔍	t3.micro	3/3 checks passed	View alarm

- Installed maria-db. through command

```
[root@ip-192-168-1-101 ~]#
[root@ip-192-168-1-101 ~]# sudo dnf install mariadb105-server -y
Amazon Linux 2023 Kernel Livepatch repository 182 kB/s | 31 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
mariadb105-server x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 10 M
Installing dependencies:
mariadb-connector-c x86_64 3.3.10-1.amzn2023.0.1 amazonlinux 211 k
mariadb-connector-c-config noarch 3.3.10-1.amzn2023.0.1 amazonlinux 9.9 k
mariadb105 x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 1.5 M
mariadb105-common x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 28 k
mariadb105-errmsg x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 212 k
mysql-selinux noarch 1.0.4-2.amzn2023.0.3 amazonlinux 36 k
perl-DBD-MariaDB x86_64 1.22-1.amzn2023.0.4 amazonlinux 153 k
perl-DBI x86_64 1.643-7.amzn2023.0.3 amazonlinux 700 k
perl-File-Copy noarch 2.34-477.amzn2023.0.7 amazonlinux 20 k
perl-Math-BigInt noarch 1:1.9998.39-2.amzn2023.0.2 amazonlinux 202 k
perl-Math-BigRat noarch 0.2624-500.amzn2023.0.2 amazonlinux 42 k
perl-Math-Complex noarch 1.59-477.amzn2023.0.7 amazonlinux 46 k
perl-Sys-Hostname x86_64 1.23-477.amzn2023.0.7 amazonlinux 16 k
Installing weak dependencies:
mariadb105-backup x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 6.0 M
mariadb105-cracklib-password-check x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 13 k
mariadb105-gssapi-server x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 15 k
```

- Checked the status of it maria-db. it is active and running

```
sr/lib/systemd/system/mariadb.service.
[root@ip-192-168-1-173 ~]# sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-04-29 12:57:54 UTC; 1min 44s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 3873 (mariabdd)
   Status: "Taking your SQL requests now..."
     Tasks: 8 (limit: 1014)
    Memory: 66.5M
       CPU: 537ms
   CGroup: /system.slice/mariadb.service
           └─3873 /usr/libexec/mariabdd --basedir=/usr

Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: The second is mysql@local
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: you need to be the system
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: After connecting you can
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: able to connect as any of
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: See the MariaDB Knowledge
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: Please report any problem
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: The latest information abo
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: Consider joining MariaDB's
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal mariadb-prepare-db-dir[3828]: https://mariadb.org/get-i
Apr 29 12:57:54 ip-192-168-1-173.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.5 data
[root@ip-192-168-1-173 ~]#
```

```
[root@ip-192-168-1-101 ~]# DBName=ec2db
[root@ip-192-168-1-101 ~]# DBPassword=admin98764
[root@ip-192-168-1-101 ~]# DBRootPassword=admin98764
[root@ip-192-168-1-101 ~]# DBUser=ec2dbuser
[root@ip-192-168-1-101 ~]# echo $DBName
ec2db
[root@ip-192-168-1-101 ~]#
```

2. Insert some dummy data.

- Added some dummy data in mysql

```
MariaDB [(none)]> CREATE DATABASE arundb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> Use arundb;
Database changed
MariaDB [arundb]> CREATE TABLE employees (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> name VARCHAR(50),
  -> role VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.008 sec)

MariaDB [arundb]> INSERT INTO employees (name, role)
  -> VALUES
  -> ('Arun', 'DevOps Engineer'),
  -> ('Sai', 'Software Engineer'),
  -> ('Mahesh', 'Tester'),
  -> ('Raju', 'Frontend');
Query OK, 4 rows affected (0.002 sec)
Records: 4 Duplicates: 0 Warnings: 0

MariaDB [arundb]> SELECT * FROM employees;
+-----+-----+-----+
| id | name | role |
+-----+-----+-----+
| 1 | Arun | DevOps Engineer |
| 2 | Sai | Software Engineer |
| 3 | Mahesh | Tester |
| 4 | Raju | Frontend |
+-----+-----+-----+
4 rows in set (0.000 sec)
```

3. Take the backup of dummy data on EC2.

- Used mysql dump to take logical backup of database from EC2.

```
[root@ip-192-168-1-101 ~]# mysqldump -u root -p arundb > arundb_backup.sql
Enter password:
[root@ip-192-168-1-101 ~]# ls -l testdb_backup.sql
-rw-r--r--. 1 root root 840 Apr 30 07:54 testdb_backup.sql
[root@ip-192-168-1-101 ~]#
```

4. Launch MariaDB RDS instance.

- Launch Maria -db. RDS instance

Aurora and RDS > Databases > Create database

Create database [Info](#)

Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☐ MySQL

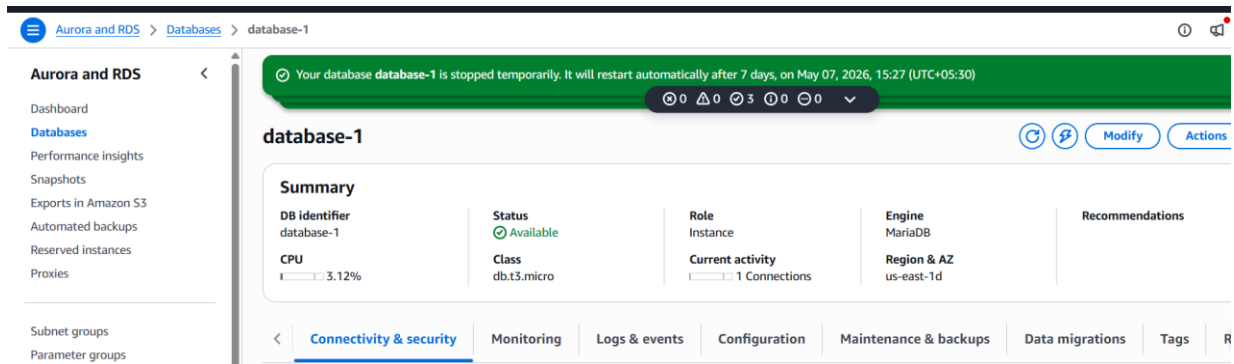
☐ PostgreSQL

☒ MariaDB

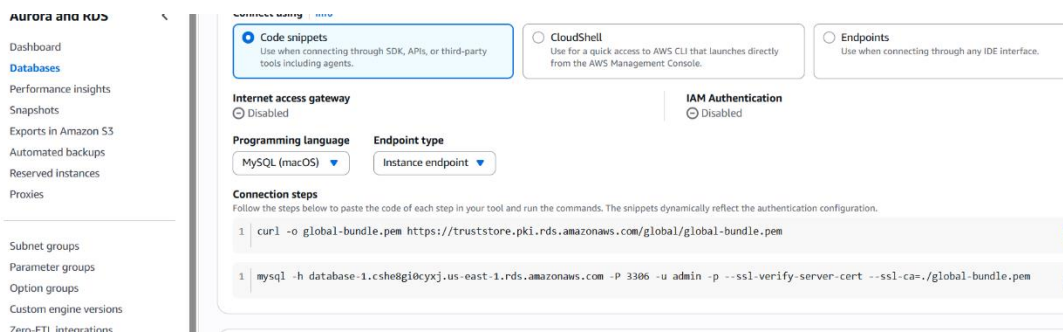
☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2



- **Maria-db. End point in RDS**



5. Migrate database from EC2 to RDS.

- **Migrated data from EC2 to RDS using SQL dump file.**

```
Enter password:
[root@ip-192-168-1-101 ~]# mysql -h database-1.cshe8gi0cyxj.us-east-1.rds.amazonaws.com -u admin -p --ssl-ca=/home/ec2-user/global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 81
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE arundb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [arundb]> SHOW TABLES;
+-----+
| Tables_in_arundb |
+-----+
| employees         |
+-----+
1 row in set (0.001 sec)

MariaDB [arundb]> SELECT * FROM employees;
+-----+
| id | name  | role           |
+-----+
| 1  | Arun  | DevOps Engineer |
| 2  | Sai   | Software Engineer |
| 3  | Mahesh | Tester         |
| 4  | Raju  | Frontend        |
+-----+
4 rows in set (0.001 sec)
```

6. Install MySQL DB on EC2.

- Installed the Mysql database on EC2.

```
lines 40-72/72 (END)
[3]+ Stopped                                sudo journalctl -xeu mysqld
[root@ip-192-168-1-101 ~]# sudo systemctl stop mysqld
[root@ip-192-168-1-101 ~]# sudo rm -rf /var/lib/mysql
[root@ip-192-168-1-101 ~]# sudo mysqld --initialize --user=mysql
[root@ip-192-168-1-101 ~]# sudo chown -R mysql:mysql /var/lib/mysql
[root@ip-192-168-1-101 ~]# sudo systemctl start mysqld
[root@ip-192-168-1-101 ~]# sudo systemctl enable mysqld
[root@ip-192-168-1-101 ~]# sudo systemctl status mysqld
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-30 11:10:20 UTC; 15s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
   Main PID: 35018 (mysqld)
   Status: "Server is operational"
    Tasks: 38 (limit: 1014)
   Memory: 378.5M
      CPU: 1.734s
   CGroup: /system.slice/mysqld.service
           └─35018 /usr/sbin/mysqld

Apr 30 11:10:18 ip-192-168-1-101.ec2.internal systemd[1]: Starting mysqld.service - MySQL Server...
Apr 30 11:10:20 ip-192-168-1-101.ec2.internal systemd[1]: Started mysqld.service - MySQL Server.
[root@ip-192-168-1-101 ~]#
```

- We can see the Mysql database

```
[root@ip-192-168-1-101 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 23
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.00 sec)
```

7. Launch MySQL RDS image.

- In RDS created database of MySql

[Aurora and RDS](#) > [Databases](#) > Create database

Create database Info

Free plan has access to limited features and resources
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[Upgrade plan](#)

Engine options

Engine type Info

☐ Aurora (MySQL Compatible)
 

☐ Aurora (PostgreSQL Compatible)
 

☒ MySQL
 

☐ PostgreSQL
 

☐ MariaDB
 

☐ Oracle
 

☐ Microsoft SQL Server
 

☐ IBM Db2
 

- **Successfully created new database for Mysql**

Successfully created database database-2 [View connection details](#) ×

You can use settings from database-2 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

database-2

[Refresh](#) [Settings](#) [Modify](#) [Actions](#)

Summary		Status	Role	Engine	Recommendations
DB identifier database-2	✓ Available	Instance	MySQL Community		
CPU 7.86%	Class db.t3.micro	Current activity 0 Connections	Region & AZ us-east-1a		

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.00 sec)

mysql>
```

8. Configure multi-AZ.

- **Multi-AZ could not be enabled due to free tier limitations. Multi-AZ requires additional cost and multiple availability zones.**

9. Take backup of DB and restore the DB.

- Taken a snapshot from mysql database 2

☰ [Aurora and RDS](#) > [Snapshots](#) > Take snapshot

Take DB Snapshot

Preferences

To take a DB Snapshot, choose a database and name your DB Snapshot.

DB instance

DB Instance identifier. This is the unique key that identifies a DB Instance.

database-2 ▼

Snapshot name

Identifier for the DB Snapshot.

Mysql-snapshot

Snapshot identifier is case insensitive, but stored as all lower-case, as in "mysnapshot". Cannot be null, empty, or blank. Must contain from 1 to 255 alphanumeric characters or hyphens. First character must be a consecutive hyphens.

- Successfully created snapshot for database 2

☰ [Aurora and RDS](#) > [Snapshots](#) > mysql-snapshot

Aurora and RDS

[Dashboard](#)

[Databases](#)

[Performance insights](#)

[Snapshots](#)

[Exports in Amazon S3](#)

[Automated backups](#)

[Reserved instances](#)

[Promotes](#)

☑ Successfully created snapshot Mysql-snapshot. [View details](#)

mysql-snapshot [Actions](#)

Details

ARN
arn:aws:rds:us-east-1:928430096503:snapshot:mysql-snapshot

Option group
default:mysql-8-4

VPC
[vpc-086492cb23d24283d](#)

Zone
us-east-1a

Status
Available

- Now restore the snapshot

☑ Successfully created snapshot Mysql-snapshot. [View details](#)

Snapshots

[Manual](#) | [System](#) | [Shared with me](#) | [Public](#) | [Backup service](#) | [Exports in Amazon S3](#)

Manual snapshots (2)

☐	Snapshot name	Engine version	DB instance or cluster
☐	database-1-snapshot	11.8.6	database-1
☒	mysql-snapshot	8.4.8	database-2

[Actions](#)

[Take snapshot](#)

Restore snapshot

Copy snapshot

Share snapshot

Migrate snapshot

Upgrade snapshot

Export to Amazon S3

- Restored the Snapshot database of mysql

Snapshot is being restored
Restoring mysql-snapshot to Mysql-restore-db. This might take several minutes.

Databases (2) Group resources ⚡ 🔄 Modify Actions ▼

🔍 Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ..
database-2	Available	Instance	MySQL Co...	SECOND	us-east-1
mysql-restore-db	Available	Instance	MySQL Co...	SECOND	us-east-1

- Snapshot was created and restored to a new RDS instance to simulate disaster recovery

```
[root@ip-192-168-1-101 ~]# mysql -h <endpoint> -u admin -p --ssl-ca=...
-bash: endpoint: No such file or directory
[root@ip-192-168-1-101 ~]# mysql -h database-2.cshe8gi0cyxj.us-east-1.rds.amazonaws.com -u admin -p --ssl-ca=/home/ec2-user/global-bundle.pem
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 558
Server version: 8.4.8 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.03 sec)
```

10. Create read replica.

- Open database and select any database and in actions click replica

Databases (2) Group resources ⚡ 🔄 Modify Actions ▲ Create

🔍 Filter by databases

DB identifier	Status	Role	Engine	Upg
database-2	Available	Instance	MySQL Co...	SEC
mysql-restore-db	Available	Instance	MySQL Co...	SEC

Stop temporarily

Reboot

Delete

Set up EC2 connection

Set up Lambda connection

Migrate data from EC2 database

Create read replica

Create Aurora read replica

Create blue/green deployment

Promote

- Created Replica of database -2

Aurora and RDS > Databases

Successfully created replica **mysql-replica** in US East (N. Virginia)

Databases (2) Group resources Modify Actions Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size
database-2	Available	Primary	MySQL Co...	SECOND	us-east-1a	db.t3.1
mysql-replica	Available	Replica	MySQL Co...	SECOND	us-east-1d	db.t3.1

- Created read replica to improve read scalability and performance.

```
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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| arundb   |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.02 sec)

mysql> use arundb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM employees;
+----+-----+-----+
| id | name  | role           |
+----+-----+-----+
| 1  | Arun  | DevOps Engineer |
| 2  | Sai   | Software Engineer |
| 3  | Mahesh | Tester         |
| 4  | Raju  | Frontend       |
+----+-----+-----+
4 rows in set (0.01 sec)
```